Round-Indexed Convergence of SCOPE-FD under Partial Participation

This note is the working derivation behind Section V-B of the manuscript. It answers R1-C1, R2-C3 and Comment 1 of the attached review, which are one question asked three times: does the FedTSKD convergence bound survive replacing full participation with SCOPE-FD's deterministic rotation? Facts 1–5 are read directly out of Mu, Garg & Ratnarajah [1]. Corollaries 1–2 compose their Theorem 1 with the manuscript's own Proposition 1. Nothing else is assumed.

Notation follows the manuscript. E_r is the number of local SGD steps in round r, E_d the number of distillation steps per round, K the cohort size, N the pool size and R the round horizon.

1 What FedTSKD assumes and proves

Fact 1. FedTSKD has no client-selection mechanism. Its Algorithm 1 loops over all N clients every round, and Section VI states this is by design: *"Full participation is considered... since the communication payload required in the FD algorithm is much less than that in the FL scheme."* SCOPE-FD's partial-participation setting extends a case the original paper left open. It does not violate an assumption that paper made.

Fact 2. Assumptions 1–4 of [1] are smoothness and strong convexity of L_n, Q_n (Assumptions 1–2) and bounded-variance and bounded-gradient conditions on their stochastic gradients (Assumptions 3–4). Assumption 3's "uniformly and independently" clause is about minibatch sampling $\mathcal{D}_{n,t} \subset \mathcal{D}_n$ *within* client n's own data, not about which clients are admitted to round r. None of the four references $\mathcal{S}_r$.

Fact 3. The proof in [1] (Lemmas 1–2, Corollaries 1–2, Theorem 1, Appendices B–F) is carried out per client, tracking a single client's weight sequence $\boldsymbol{w}_{n,t}$ through $\boldsymbol{w}_{n,t+1} = \boldsymbol{w}_{n,t} - \eta_t \nabla L_n(\boldsymbol{w}_{n,t}, \mathcal{D}_{n,t})$ and its distillation analogue. No step depends on any other client. This follows from Fact 1: with no selection mechanism in the algorithm, there is nothing selection-related for the proof to condition on.

Fact 4. The time index t ranges over $\mathcal{T}_{LT} \cup \mathcal{T}_{SD} \cup \mathcal{T}_{LD}$, client n's own elapsed local-training and distillation steps. Under full participation this coincides with the round counter, since every client passes through every phase every round. Under partial participation it does not: a client's t only advances on rounds it is selected. This is the one place a new statement is needed, not a violated assumption.

Fact 5. The problem statement in [1] is N separate per-client minimizations, $\boldsymbol{w}^* = \arg\min_{\boldsymbol{w}} L_n(\boldsymbol{w}),\ \forall n$. The manuscript's additive objective $F(\{\boldsymbol{w}_n\}) = \sum_n \lambda_n L_n(\boldsymbol{w}_n)$ (its Eq. 2) inherits this directly, so $F^* = \sum_n \lambda_n L_n^*$ is not a simplification introduced here.

2 Given

Proposition 1 (manuscript, already proved): under SCOPE-FD with $\alpha_u + \alpha_d < 1$,

$$\left| n_i^{(R)} - n_j^{(R)} \right| \le 1 \quad \forall i, j, R. \tag{1}$$

Its exact-count refinement (manuscript Eq. 17), with $m = KR \bmod N$, puts every client at

$$n_i^{(R)} \;\geq\; \left\lfloor \frac{KR}{N} \right\rfloor \;\geq\; \frac{KR}{N} - 1. \tag{2}$$

FedTSKD's per-client bound ([1], Theorem 1):

$$\mathbb{E}\big[L_n(\boldsymbol{w}_{n,t+1})\big] - L_n^* \leq \frac{\psi}{2} \cdot \frac{\nu_n}{t+1+\beta_1}, \tag{3}$$

with t as in Fact 4 and ψ, ν_n, β_1 as defined there.

3 Why the original Remark understated its case

The manuscript's earlier Remark (Section V-B) stated that the FedTSKD bound "continues to hold... without modification." At the level of (3) this is nearly a tautology: by Facts 1–3 the bound was never conditioned on a selection mechanism, so restating that it still holds needs no argument. This is R1's objection, and it is correct. The Remark's analytical content was minimal because it was never really about partial participation.

The question that does have content is how the bound behaves in *communication rounds* R, not in t. Under full participation the two coincide. Under SCOPE-FD they do not, since t only advances on rounds a client is selected (Fact 4), and (3) alone bounds nothing in terms of R. Proposition 1 is what closes that gap, for every client at once.

4 The round-indexed corollaries

Let $E_{\min} = \min_{r \leq R} E_r > 0$ and write $\Gamma_R = (E_{\min} + E_d)\big(KR/N - 1\big) + \beta_1$. Both E_r and E_d are fixed by the training schedule and are independent of the selection.

Corollary 1 (Per-client rate in rounds)**.** *Under Assumptions 1–4 of [1] and $\alpha_u + \alpha_d < 1$, for every R large enough that $\Gamma_R > 0$, every client n satisfies*

$$\mathbb{E}\big[L_n(\boldsymbol{w}_n^{(R)})\big] - L_n^* \;\leq\; \frac{\psi\,\nu_n}{2\,\Gamma_R} \;=\; \mathcal{O}\!\left(\frac{N}{KR}\right). \tag{4}$$

Proof. Only $n \in \mathcal{S}_r$ update in round r, so client n's counter t advances only on rounds it participates in, by $E_r + E_d \geq E_{\min} + E_d$ steps each time. Hence $t_n^{(R)} \geq (E_{\min} + E_d)\, n_n^{(R)}$. Substituting into (3) at $t + 1 = t_n^{(R)}$ and applying (2), which holds for every client simultaneously, gives (4). □

The word *every* is the content of this corollary. No policy without Proposition 1's balance guarantee can make the same claim uniformly, since an in-expectation fairness property alone permits some client's realized $n_i^{(R)}$ to fall arbitrarily below KR/N at any finite R.

Corollary 2 (Global objective, round-indexed)**.** *Under the same hypotheses, with $\lambda_n = |\mathcal{D}_n|/|\mathcal{D}|$ and $\bar{\nu} = \sum_n \lambda_n \nu_n$,*

$$\mathbb{E}\big[F(\{\boldsymbol{w}_n^{(R)}\})\big] - F^* \;\leq\; \frac{\psi\,\bar{\nu}}{2\,\Gamma_R} \;=\; \mathcal{O}\!\left(\frac{N}{KR}\right). \tag{5}$$

Proof. Γ_R does not depend on n, so weighting (4) by λ_n and summing gives (5). Use Fact 5 for $F^* = \sum_n \lambda_n L_n^*$. □

Remark 1 (Consistency check)**.** *At $K = N$, (5) gives $\mathcal{O}(1/R)$, matching FedTSKD's original rate. This must hold: at full participation SCOPE-FD's selector has nothing to do, so the bound has to degenerate back to [1]'s own rate.*

Remark 2 (What is not claimed)**.** *Corollary 2 does not beat uniform random selection in order. Both are $\mathcal{O}(N/(KR))$. The gain of the deterministic rotation is that the bound holds for each client separately at every finite R rather than only on average, which is the same distinction Proposition 1 draws for the participation counts themselves.*

5 Assumption audit

(a) Smoothness and convexity (Assumptions 1–2): properties of L_n, Q_n alone. SCOPE-FD changes no loss function. Unchanged.

(b) Bounded variance and gradient (Assumptions 3–4): the same, plus within-client minibatch sampling, which is unrelated to selecting $\mathcal{S}_r$. Unchanged.

(c) Aggregation: the proof in [1] uses $\mathcal{S}_r$ only through the manuscript's aggregation rule (Eq. 5), and SCOPE-FD uses the identical rule. Unchanged.

(d) Sampling scheme: Theorem 1 of [1] is proved under full participation with no selection mechanism and no step referencing any client but n itself (Facts 1, 3). There is no sampling assumption for a deterministic rule to satisfy or violate. Corollary 2 is therefore an unconditional corollary of Theorem 1 composed with Proposition 1, not a hedged parallel result.

6 Status

Corollaries 1–2, the consistency check and the assumption audit are now in Section V-B of the manuscript, and the three reviewer replies that depended on them are written.

Nothing further is outstanding. Corollaries 1 and 2 are unconditional given Assumptions 1–4 of [1] and $\alpha_u + \alpha_d < 1$, so they need no experiment to support them. A full-participation reference run at K=N=30 would let the empirical log-log exponent of the K sweep be reported against the -1 that Corollary 2 predicts, but that is a confirmation of a bound already proved rather than evidence the bound depends on, and the submitted version does not rely on it.

References

[1] Y. Mu, N. Garg, and T. Ratnarajah, "Federated Distillation in Massive MIMO Networks: Dynamic Training, Convergence Analysis, and Communication Channel-Aware Learning," *IEEE Trans. Cogn. Commun. Netw.*, vol. 10, no. 4, pp. 1535–1550, Aug. 2024. doi: 10.1109/TCCN.2024.3378215.